Optimal Subsidy Design in Natural Disaster Insurance Markets: Evidence from NFIP

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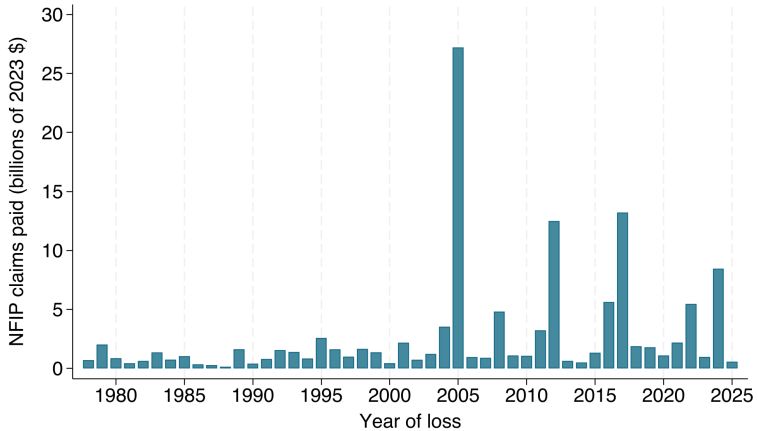
NYU Stern

Motivation and setting

Motivation

- Natural disasters cause large economic damages, and losses will grow as the climate changes
  - Flooding is the costliest: $32 billion in expected annual losses, projected to rise 26% by 2050 (Wing et al. 2022)
  - Counting indirect costs, flooding costs the U.S. $180–500 billion a year (Joint Economic Committee 2024)
- Adapting to rising risk requires costly investments in the housing stock
- These investments are likely under-provided, in part because subsidized insurance mutes their return
- Incentivizing adaptation is a second-best problem: efficiency against affordability

Motivation: Flood Losses Are Large and Lumpy



Source: OpenFEMA NFIP redacted claims, all policies, by year of loss, 2023 dollars.

Setting

Insurance: National Flood Insurance Program (NFIP)

- Primary insurer against flood risk in the U.S.
  - 4.7 million policies and $1.3 trillion in coverage in 2025; the largest natural disaster insurance program in the world
- Premiums held below risk by statute, on affordability grounds: before Risk Rating 2.0 the average policy paid about half its full-risk premium (Gruber & Solomon 2026)
- Ongoing reform is moving premiums toward risk

Adaptation: Home elevations

- Primary property-level protection against flood risk
- Mandatory for new construction in high-risk zones
- Retrofitting an existing house is effective but costly: median declared permit cost $146,000; FEMA grant projects $160,000–225,000 per home

Research Question

- **Question:** A public insurer can subsidize premiums or adaptation investment. What is the optimal policy design?
- **Key forces:**
  - Premium subsidies keep insurance affordable but mute the return to investment
  - Risk-based pricing restores the incentive but pushes households out of coverage
  - Investment subsidies keep households insured and lower claims directly
- **This paper:** joint model of insurance take-up and investment, estimated on NFIP microdata

Related Literature

1. Subsidized disaster insurance and adaptation

- Low willingness to pay; houses built under elevation mandates less likely to insure (Wagner 2022; Hovekamp & Wagner 2023)
- Premium subsidies remain optimal once the spillover onto disaster aid is accounted for (Gruber & Solomon 2026)
- Subsidies slow long-run adaptation; targeting restores it (Obolensky 2024)

2. Preventive care in health insurance

- Insurers should cover prevention when the insured ignore its effect on premiums (Ellis & Manning 2007; Newhouse 2006)
- Cost-sharing on upstream care raises downstream claims, often paid by another payer (Chandra, Gruber & McKnight 2010; Starc & Town 2018)
- Root: market insurance and self-protection as complements (Ehrlich & Becker 1972)

Related Literature (Continued)

3. The disaster insurance puzzle: why is take-up so low?

- Flood insurance coverage surprisingly low (Kunreuther & Pauly 2004; Wagner 2022)
- Risk misperception and salience explain take-up patterns (Gallagher 2014; Kousky 2010; Xu & Box-Couillard 2024)

4. Ex ante insurance vs. ex post aid (Samaritan's dilemma)

- Expected relief crowds out insurance demand for floods (Raschky et al. 2013; Kousky et al. 2018; Gruber & Solomon 2026) and crops (Deryugina & Kirwan 2018; Philippi & Schiller 2024)

5. Moral hazard in natural disaster insurance markets

- Location margin: insurance stimulates development and migration (Fabian 2024; Peralta & Scott 2024; Browne et al. 2019)
- Mitigation margin: small effects of insurance prices on risk reduction (Hudson et al. 2017; Kousky 2019; Gruber & Solomon 2026)

Background

The National Flood Insurance Program

- **FEMA-administered, federally backed:** over 90% of U.S. residential flood policies; cumulative claims have left the program $20.5 billion in debt to the Treasury
- **Flood Insurance Rate Maps (FIRMs)** define flood zones and the Base Flood Elevation (BFE)
  - Post-FIRM construction in high-risk zones (SFHAs) must be elevated above the BFE
  - Pre-FIRM homes are exempt from elevation requirements and were charged heavily subsidized premiums
- **Pricing reforms:**
  - **Biggert-Waters (2012) / HFIAA (2014):** phase out pre-FIRM subsidies
  - **Risk Rating 2.0 (2021–23):** replaces zone-based rating with property-level full-risk premiums
  - Under both, subsidized policies converge to full-risk prices along a glide path of 5–18% annual increases

Home Elevations

- **Cheap ex ante, expensive ex post.** Building a new house a foot higher adds about 0.25–1.5% to construction cost; lifting an existing house costs $150,000–225,000
- **What drives retrofit cost:** lifting the slab foundation, the size and weight of the house, and how high it has to go
- **FEMA elevation grants** (Hazard Mitigation Assistance):
  - **HMGP**: post-disaster, administered by the state emergency agency; about 8,900 single-family elevations and $2.2 billion since 1993
  - **FMA**: financed out of NFIP premiums, targets repetitive-loss homes; about 5,300 elevations and $1.0 billion since 1997, roughly 200 homes a year
- **Elevations are modest:** in the past 20 years FMA has elevated about 3% of roughly 181,000 repetitive-loss properties on NFIP's list

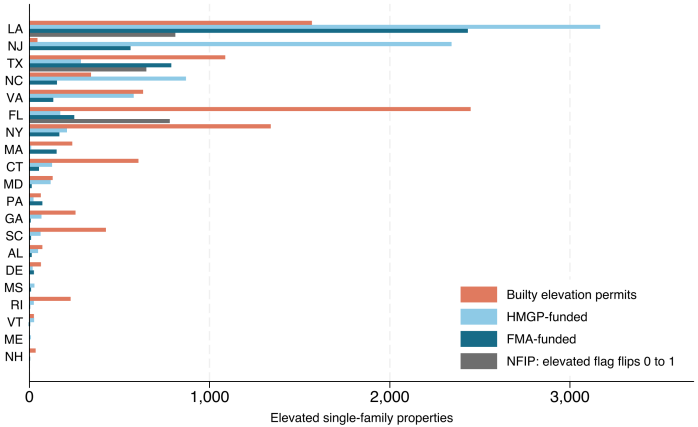
Data

Data Sources

Sample: NFIP-insured single-family homes in SFHAs, Florida, Louisiana, Texas, 2009–2025

- **NFIP.** Universe of policy transactions, plus the claims and multiple-loss files
  - Premiums, coverage, flood zone, post-FIRM status, RL/SRL status, claims by year
  - Addresses are redacted: properties identified by construction date, block group, and policy origination date
- **FEMA HMA.** Elevation grants: program (FMA vs. HMGP), cost, and number of properties, merged at the county
- **Builty permits.** Building permits from municipal feeds Appendix
  - Elevations identified from permit text; retrofits separated from elevated new construction; declared project value
- **ATTOM tax assessor.** Property-year market value and characteristics; carries the address that links permits to NFIP
- **NFHL flood maps.** FEMA flood zone for ATTOM properties

Why These Three States



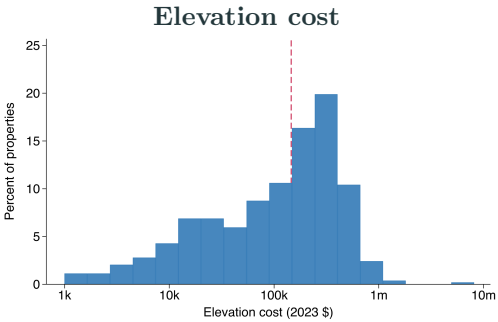
- Counts by source across 20 states; HMA counts funded projects only
- Louisiana is large on both the permit and the grant side; Florida has the most permits; Texas is in between
- On the analysis sample: 3.0 (LA), 2.3 (TX) and 1.1 (FL) elevation events per 1,000 insured SFHA homes

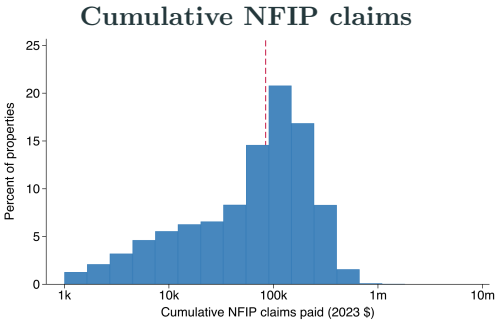
Summary Statistics

NFIP-insured single-family properties in SFHAs, FL LA TX, 2009–2025

	All	Pre-FIRM	Post-FIRM
Annual premium, median ($)	644	888	574
Claims paid / premiums collected	1.07	1.26	0.82
Any claim in a year (share)	0.012	0.016	0.008
Cumulative claims, median ($)			
among claimants	82,600	96,500	45,900
among repetitive loss	104,400	112,800	72,500
among severe repetitive loss	122,300	124,600	98,200
Property value, median ($)	242,300	220,900	261,300
Elevated ex ante	450,281	173,018	277,263
Elevation retrofit	4,993	3,543	1,450
Retrofit cost, median ($)	145,900	159,000	43,000
Number of properties	3,291,652	1,621,829	1,669,823

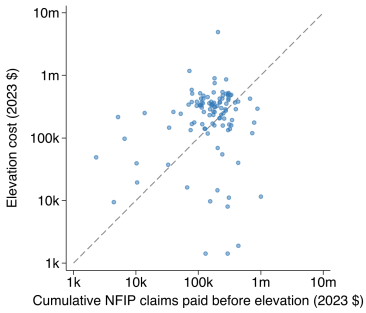
Elevation Costs vs. Cumulative Claims





- The two distributions overlap: for many claimants, the claims already paid are of the order of an elevation
- Wide dispersion on both sides, so the return to elevating differs a lot across homes

Elevation Cost Against Claims Paid Before the Permit



- One point per elevated home with a declared cost and prior claims
- Below the 45-degree line, the claims already paid exceed the cost of elevating
- Most homes elevate at a cost several times their prior claims; for a minority, one flood had already cost more than the elevation

Reduced-Form Evidence

Event-Study Design

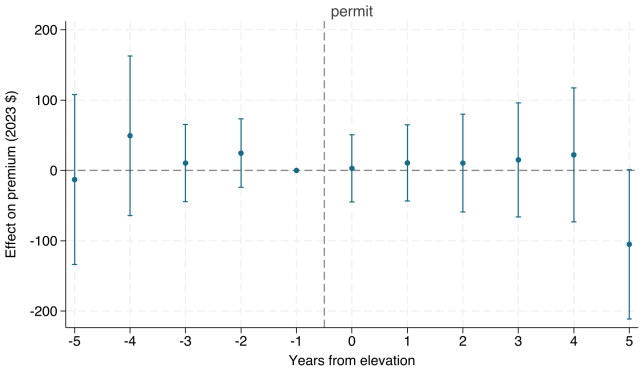
For home i in county c, flood zone z, and policy year t, with an elevation permit in year τ_i:

$$y_{it} \;=\; \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \, \mathbf{1}\{t - \tau_i = k\} \;+\; \alpha_i \;+\; \gamma_{czt} \;+\; \varepsilon_{it}$$

- y_{it}: annual premium, or paid claim, in 2023 dollars
- α_i: home fixed effect. γ_{czt}: county × flood zone (A/V) × year, which absorbs local floods and the move to risk-based pricing
- β_k: change relative to the year before the permit; endpoints binned at ± 5. Each β_k is the cohort-share-weighted average of cohort-specific coefficients (Sun & Abraham 2021)
- Treated: homes with a Builty elevation permit in 2012–2022. Controls: homes never elevated by any source, so staggered permit timing does not contaminate β_k
- Standard errors clustered by county

Identifying assumption: absent the permit, outcomes of permitted homes would have moved in parallel with never-elevated homes in the same county, zone and year. Caveat: permits follow floods, so $k = -1$ is usually a flood year

Elevating Does Not Lower the Premium



- The insurer does not reward the investment: no drop in the premium in the five years after the permit

Same-Flood Design

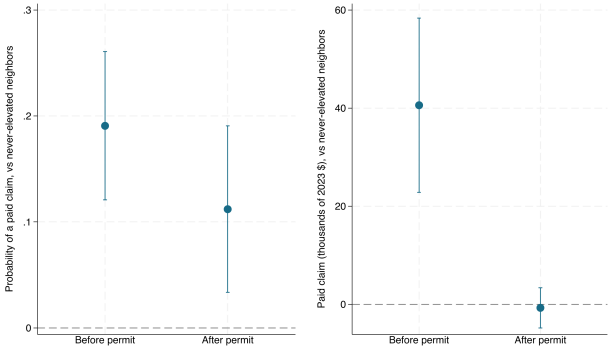
Floods are rare inside a five-year window, so effectiveness is measured on homes hit by the same flood: county-years in which at least 5% of insured homes, and at least 50 homes, have a paid claim

$$y_{it} \;=\; \beta^{\text{pre}}\,\text{Pre}_{it} \;+\; \beta^{\text{post}}\,\text{Post}_{it} \;+\; \gamma_{czt} \;+\; \delta_{d(i)f(i)} \;+\; \varepsilon_{it}$$

- Pre_{it}, Post_{it}: permitted home observed before, or after, its permit. Omitted group: never-elevated homes
- γ_{czt}: county × flood zone × year, so the comparison is within the same flood. δ_{df}: construction decade × post-FIRM status
- β^{pre} measures selection into elevation; $\beta^{\text{pre}} - \beta^{\text{post}}$ is the damage avoided per flood
- Standard errors clustered by county

Identifying assumption: within a flood, a permitted home's excess damage over its neighbors would have stayed at its pre-permit level without the elevation

Elevating Lowers Claims



- Same flood, same county and zone: homes that later elevate claim $41k more than their neighbors before the permit, and the same as their neighbors after it
- $41k per flood year × 7% flood-year frequency ≈ $2,900 a year; NPV over 30 years at 7% ≈ $36,000, a quarter of the median cost of $146,000

Model & Empirical Strategy

Model

Joint model of insurance take-up and mitigation investment

- Household with house value v and flood risk π chooses whether to insure and whether to elevate
- Elevation cuts expected damage and, under risk-based pricing, the premium; it costs $(1 - s)\, c$ plus a non-monetary cost κ (time, disruption, the look of a raised house)
- Insurance pays capped claims at premium $p(E)$; uninsured losses are partly covered by ex post aid
- **Two policy instruments:** the premium schedule and the elevation subsidy s
- **Distortion:** subsidized premiums mute the return to elevating; risk-based premiums restore it but reduce take-up
- **Counterfactuals**, same government outlay: actuarially fair premiums vs. an elevation subsidy

Estimation

- **Cost c:** Builty declared permit cost; HMA project cost per home
- **Response of investment to price:** FMA cost shares (75/90/100%) and the elevation discount before vs. after Risk Rating 2.0
- **Effectiveness of elevation:** same-flood comparison above; literature (Hovekamp & Wagner 2023; USACE 2021)
- **Insurance demand:** take-up response to the premium reforms below
- **Risk π:** claims history and flood zone
- **Non-monetary cost κ:** residual that matches observed elevation rates in FL, LA, TX

Identifying Variation in Premiums

Two NFIP pricing reforms generate house-level variation in premiums:

- **Biggert-Waters (2012) / HFIAA (2014):** phase-out of historically subsidized premiums for pre-FIRM homes
  - Exogenous price changes differentially affect **non-adapted** properties Wagner (2022)
- **Risk Rating 2.0 (2021–):** property-level, risk-based pricing replaces the old rating system
  - Within-property price increases toward a full-risk premium Gruber and Solomon (2026)

Next Steps

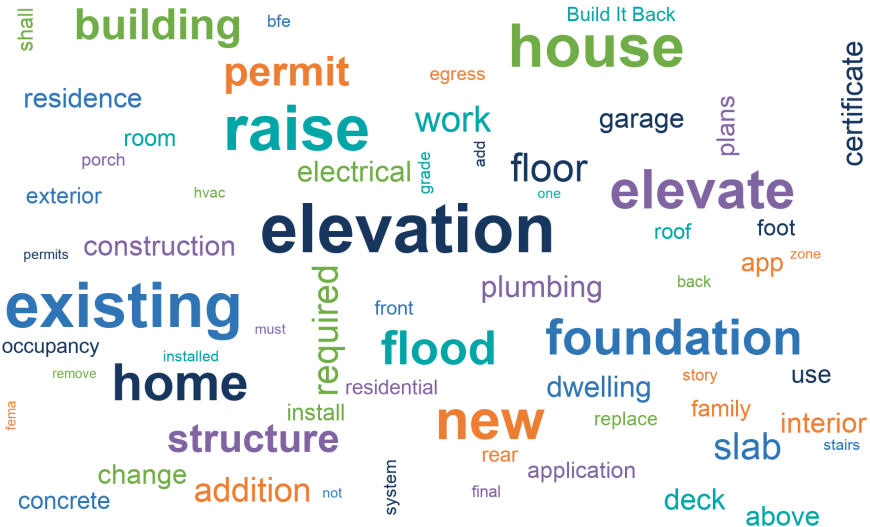
Next Steps

- **Data:** property-level grant records from Louisiana (GOHSEP, due end of October) and Rhode Island; ATTOM permit data as a replacement for Builty if the quote allows
- **Empirics:** elevation discount and take-up around Risk Rating 2.0; investment response to the FMA cost share
- **Model:** write down the household's elevation and insurance choice and what each moment pins down

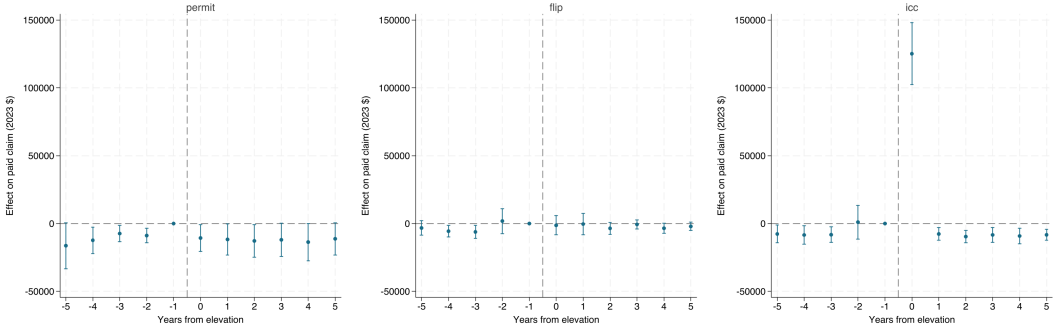
Appendix

Builty Word Cloud

Back to Data Sources

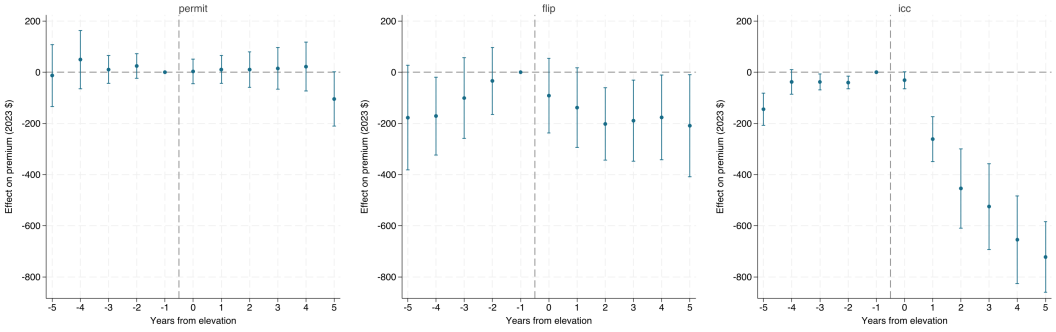


Appendix: Claims Around the Elevation, by Source



- The permit follows the flood (year −1, the reference); the ICC year is the flood itself (year 0); a flag flip is the insurer's re-rating, years after the work
- For no source do claims fall below their normal pre-flood level after the event: the within-home design compares average years, and the return to elevating appears only in flood years

Appendix: Premiums Around the Elevation, by Source



- Permitted homes: no change. Flag-flip and ICC homes: the premium falls after the event, with a downward pre-trend, consistent with the insurer re-rating the house rather than a return to a private investment

Appendix: Federal Spending on Home Elevations

